Reading Across Columns: A Local Alignment Demonstration

A reproducible sample for checking reading order, bilingual navigation and persistent source anchors.

1. Reading order

L1. A two-column page should be read from the top of the left column to its bottom, followed by the right column. Sorting every line by vertical position alone interleaves unrelated passages.

L2. Each paragraph keeps its original page number and a set of line rectangles. The reader uses these coordinates to highlight exactly where the passage appeared in the source PDF.

L3. A translation is linked to the paragraph identity rather than its screen position. Changing the zoom level or making the Chinese text longer must not move the original source anchor.

2. A verifiable translation

R1. The Chinese text in this demonstration was written for the sample. It is not an output from an external model. Real API translations are requested only after the user explicitly starts a translation task.

R2. Numbers, variable names and citation markers must remain traceable. For example, the value 0.25 and the reference [3] should survive translation without silent changes or invented supporting evidence.

R3. When a parser mistakes diagram labels for prose, the user can exclude those fragments. Manual corrections are stored locally and the remaining paragraphs keep their source locations.

Demonstration note: this document contains no experimental results. The second page tests a full-width section between two groups of columns.

3. Mixed layout, first band

A1. A page may begin with two columns and then insert a full-width section. The first left passage must come before the first right passage, even if their lines have matching vertical coordinates.

4. First right passage

A2. After both columns above the separator have been read, the full-width passage becomes the next item. It should not be inserted midway through a paragraph from either column.

5. Full-width section: a reading-order boundary

W1. This passage separates the upper and lower column groups. Its source rectangle spans the central gutter.

6. Lower left passage

B1. Reading now resumes at the lower left column. Stable block identities let the application keep this passage linked to its Chinese translation after the paragraph order is corrected.

B2. Local persistence means that translations and manual exclusions remain available after closing the browser. The original PDF remains an unchanged source file.

7. Lower right passage

B3. A scanned page has no reliable text layer. This prototype reports that limitation instead of pretending that a blank extraction is a successful translation. The page image remains readable.

B4. To test the full workflow, configure a compatible API, select one paragraph and request a translation. The returned text should be marked with its actual model name and remain linked to this PDF region.